



June 13, 2025

Brainlab AG
Sadwini Suresh
QM Consultant
Olof-Palme-Str.9
Munich, 81829
Germany

Re: K243633

Trade/Device Name: Brainlab Elements (7.0);
Brainlab Elements Image Fusion (5.0);
Brainlab Elements Image Fusion Angio (1.0);
Brainlab Elements Contouring (5.0);
Brainlab Elements Fibertracking (3.0);
Brainlab Elements BOLD MRI Mapping (1.0)

Regulation Number: 21 CFR 892.2050

Regulation Name: Medical image management and processing system

Regulatory Class: Class II

Product Code: QIH, JAK, LLZ

Dated: May 27, 2025

Received: May 27, 2025

Dear Sadwini Suresh:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of

Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device" (<https://www.fda.gov/media/99812/download>) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (<https://www.fda.gov/media/99785/download>).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at <https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system>.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the

Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

A handwritten signature in black ink, appearing to read 'D. Krainak', is written over a large, light blue, semi-transparent 'FDA' watermark.

Daniel M. Krainak, Ph.D.

Assistant Director

DHT8C: Division of Radiological Imaging
and Radiation Therapy Devices

OHT8: Office of Radiological Health

Office of Product Evaluation and Quality

Center for Devices and Radiological Health

Enclosure

Indications for Use

Submission Number (if known)

K243633

Device Name

Brainlab Elements (7.0);
Brainlab Elements Image Fusion (5.0);
Brainlab Elements Image Fusion Angio (1.0);
Brainlab Elements Contouring (5.0);
Brainlab Elements Fibertracking (3.0);
Brainlab Elements BOLD MRI Mapping (1.0)

Indications for Use (Describe)

Brainlab Elements Image Fusion is an application for the co-registration of image data within medical procedures by using rigid and deformable registration methods. It is intended to align anatomical structures between data sets. It is not intended for diagnostic purposes.

Brainlab Elements Image Fusion is indicated for planning of cranial and extracranial surgical treatments and preplanning of cranial and extracranial radiotherapy treatments.

Brainlab Elements Image Fusion Angio is a software application that is intended to be used for the co-registration of cerebrovascular image data. It is not intended for diagnostic purposes.

Brainlab Elements Image Fusion Angio is indicated for planning of cranial surgical treatments and preplanning of cranial radiotherapy treatments.

Brainlab Elements Fibertracking is an application for the processing and visualization of cranial white matter tracts based on Diffusion Weighted Imaging (DWI) data for use in treatment planning procedures. It is not intended for diagnostic purposes.

Brainlab Elements Fibertracking is indicated for planning of cranial surgical treatments and preplanning of cranial radiotherapy treatments.

Brainlab Elements Contouring provides an interface with tools and views to outline, refine, combine and manipulate structures in patient image data. It is not intended for diagnostic purposes.

Brainlab Elements Contouring is indicated for planning of cranial and extracranial surgical treatments and preplanning of cranial and extracranial radiotherapy treatments.

Brainlab Elements BOLD MRI Mapping provides tools to analyze blood oxygen level dependent data (BOLD MRI Data) to visualize the activation signal. It is not intended for diagnostic purposes.

Brainlab Elements BOLD MRI Mapping is indicated for planning of cranial surgical treatments.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

510(k) Summary

June 11, 2025

General Information	
Manufacturer	Brainlab AG; Olof-Palme Str.9; 81829, Munich, Germany
Establishment Registration	8043933
Trade Name	Brainlab Elements (7.0); Brainlab Elements Image Fusion (5.0); Brainlab Elements Image Fusion Angio (1.0); Brainlab Elements Contouring (5.0); Brainlab Elements Fibertracking (3.0); Brainlab Elements BOLD MRI Mapping (1.0)
Classification Name	Automated Radiological Image Processing Software
Product Code	QIH; JAK; LLZ
Regulation Number	892.2050
Regulatory Class	II
Panel	Radiology
Predicate Device	K223106; Brainlab Elements 6.0
Reference Device	K212397; StealthStation S8 Cranial v2.0
Contact Information	
Primary Contact	Alternate Contact
Sadwini Suresh QM Consultant Phone: +49 89 99 15 68 0 Email: regulatory.affairs@brainlab.com	Chiara Cunico Senior Manager Regulatory Affairs Phone: +49 89 99 15 68 0 Email: chiara.cunico@brainlab.com

1. Indications for Use

Brainlab Elements Image Fusion is an application for the co-registration of image data within medical procedures by using rigid and deformable registration methods. It is intended to align anatomical structures between data sets. It is not intended for diagnostic purposes.

Brainlab Elements Image Fusion is indicated for planning of cranial and extracranial surgical treatments and preplanning of cranial and extracranial radiotherapy treatments.

Brainlab Elements Image Fusion Angio is a software application that is intended to be used for the co-registration of cerebrovascular image data. It is not intended for diagnostic purposes.

Brainlab Elements Image Fusion Angio is indicated for planning of cranial surgical treatments and preplanning of cranial radiotherapy treatments.

Brainlab Elements Fibertracking is an application for the processing and visualization of cranial white matter tracts based on Diffusion Weighted Imaging (DWI) data for use in treatment planning procedures. It is not intended for diagnostic purposes.

Brainlab Elements Fibertracking is indicated for planning of cranial surgical treatments and preplanning of cranial radiotherapy treatments.

Brainlab Elements Contouring provides an interface with tools and views to outline, refine, combine and manipulate structures in patient image data. It is not intended for diagnostic purposes.

Brainlab Elements Contouring is indicated for planning of cranial and extracranial surgical treatments and preplanning of cranial and extracranial radiotherapy treatments.

Brainlab Elements BOLD MRI Mapping provides tools to analyze blood oxygen level dependent data (BOLD MRI Data) to visualize the activation signal. It is not intended for diagnostic purposes.

Brainlab Elements BOLD MRI Mapping is indicated for planning of cranial surgical treatments.

2. Device Description

The Brainlab Elements are applications and background services for processing of medical images including functionalities such as data transfer, image co-registration, image segmentation, contouring and other image processing.

They consist of the following software applications:

1. Image Fusion 5.0
2. Image Fusion Angio 1.0
3. Contouring 5.0
4. BOLD MRI Mapping 1.0
5. Fibertracking 3.0

This device is a successor of the Predicate Device Brainlab Elements 6.0 (K223106).

Brainlab Elements Image Fusion is an application for the co-registration of image data within medical procedures by using rigid and deformable registration methods.

Brainlab Elements Image Fusion Angio is a software application that is intended to be used for the co-registration of cerebrovascular image data. It allows co-registration of 2D digital subtraction angiography images to 3D vascular images in order to combine flow and location information. In

particular, 2D DSA (digital subtraction angiography) sequences can be fused to MRA, CTA and 3D DSA sequences.

Brainlab Elements Contouring provides an interface with tools and views to outline, refine, combine and manipulate structures in patient image data. The output is saved as 3D DICOM segmentation object and can be used for further processing and treatment planning.

BOLD MRI Mapping provides methods to analyze task-based (block-design) functional magnet resonance images (fMRI). It provides a user interface with tools and views in order to visualize activation maps and generate 3D objects that can be used for further treatment planning.

Brainlab Elements Fibertracking is an application for the processing and visualization of information based upon Diffusion Weighted Imaging (DWI) data, i.e. to calculate and visualize cranial white matter tracts in selected regions of interest, which can be used for treatment planning procedures.

3. Substantial Equivalence

Predicate Number	Name
Subject Device	Brainlab Elements, Brainlab Elements Contouring (5.0), Brainlab Elements Fibertracking (3.0), Brainlab Elements Image Fusion (5.0), Brainlab Elements Image Fusion Angio (1.0), Brainlab Elements BOLD MRI Mapping (1.0)
Predicate device (K223106)	Brainlab Elements, Brainlab Elements Contouring (4.5), Brainlab Elements Fibertracking (2.0), Brainlab Elements Image Fusion (4.5), Brainlab Elements Image Fusion Angio (1.0), Brainlab Elements BOLD MRI Mapping (1.0)
Reference device (K212397)	StealthStation S8 Cranial v2.0

Image Fusion 5.0

Topic/ Feature	Primary Predicate Device (Brainlab Elements Image Fusion 4.5 (K223106))	Subject Device (Brainlab Elements Image Fusion 5.0)
Indications for use	Brainlab Elements Image Fusion is an application for the co-registration of image data within medical procedures by using rigid and deformable registration methods. It is	Brainlab Elements Image Fusion is an application for the co-registration of image data within medical procedures by using rigid and deformable registration methods. It is intended to align

Topic/ Feature	Primary Predicate Device (Brainlab Elements Image Fusion 4.5 (K223106))	Subject Device (Brainlab Elements Image Fusion 5.0)
	intended to align anatomical structures between data sets. The device itself does not have clinical indications.	anatomical structures between data sets. It is not intended for diagnostic purposes. Brainlab Elements Image Fusion is indicated for planning of cranial and extracranial surgical treatments and preplanning of cranial and extracranial radiotherapy treatments.
Operator profile	The intended users are medical professionals. Typical users are: - Image Guided Surgery (IGS): Neurosurgeons, Ear-Nose-Throat (ENT) surgeons and Cranio-Maxillofacial (CMF) surgeons including their assistants Radiotherapy (RT): medical professionals who perform radiation treatment planning (medical physicists, radiation oncologists, dosimetrists, physicians, etc.)	The intended users are medical professionals. Typical users are: - Image Guided Surgery (IGS): Surgeons trained in the areas of neurosurgery, spine and trauma surgery, ear-nose-throat (ENT) surgery and craniomaxillofacial (CMF) surgery including their assistants. Radiotherapy (RT): medical professionals who perform radiation treatment planning (medical physicists, radiation oncologists, dosimetrists, physicians, etc.).
Patient population	In general, there are no demographic, regional or cultural limitations for patients. It is up to the user to decide if the system shall be used to assist a certain procedure.	There are no demographic, regional or cultural limitations for patients.
Conditions of use	The system shall be used in a hospital office environment or rooms appropriate for surgical interventions or radiotherapy planning.	The system shall be used in a hospital office environment or rooms appropriate for surgical interventions or radiotherapy planning.
Computer hardware requirements	Brainlab Elements can be used on hardware that fulfills the defined minimum requirements: - Operating System: Windows 8.1 64bit - Minimum 4 logical cores - Minimum RAM: 6 GB - Graphics: Direct X compatible - Display Resolution: 1920 x 1080 (Full HD)	All platforms which fulfill the minimum requirements: IGS Workstation: - Operating system: Windows 10 - minimum 4 physical cores - RAM: 8 GB - Graphics: DirectX 12 compatible - Display resolution: 1920 x 1080 (Full HD) IGS Server / Virtual Machine - Operating System: Windows Server 2016 - RAM 16GB RT Workstation: - Operating System: Windows 10 - minimum 6 physical cores - RAM: 48GB - Graphics card: DirectX 12 - Display resolution: 1920 x 1080 (Full HD) RT Server / Virtual Machine - Operating System: Windows Server 2016 - minimum 12 physical cores

Topic/ Feature	Primary Predicate Device (Brainlab Elements Image Fusion 4.5 (K223106))	Subject Device (Brainlab Elements Image Fusion 5.0)
		- RAM: 64GB (+16GB per additional user)
Operating system	- Windows Server 2012 R2 / 2016 / 2019 - Windows 8.1 Windows 10	- Windows Server 2016 - Windows Server 2019 - Windows Server 2022 - Windows 10
Input Devices	Touch and mouse/keyboard control Touch and mouse/keyboard control	Touch and mouse/keyboard control Touch and mouse/keyboard control
Image Data Sets Format	DICOM 3D imaging Modalities, e.g. CT/XT, MRI, NM/PET, OT, are supported for automatic fusion. Other modalities are supported for manual fusion (alignment)	DICOM 3D imaging Modalities, e.g. CT/XT, MRI, NM/PET, OT, are supported for automatic fusion. Other modalities are supported for manual fusion (alignment)
Rigid co- registration, rigid image fusion	Automatic rigid co-registration (rigid fusion) of 3D DICOM image data: CT (incl. low-dose CT), MR, OT (e.g. Perfusion), MR-DTI, MR-BOLD, NM/PET, US (ultrasound). Manual rigid co-registration (rigid fusion) of other (unknown) 3D DICOM image data.	Automatic rigid co-registration (rigid fusion) of 3D DICOM image data: CT (incl. low-dose CT), MR, OT (e.g. Perfusion), MR-DWI, MR-BOLD, NM/PET, US (ultrasound). Manual rigid co-registration (rigid fusion) of other (unknown) 3D DICOM image data.
Application View Layout and Views	GUI Technology: HTML Single view layout: - Axial view is displayed by default - The user is able to switch to coronal and sagittal views - A toolbar is available on the right side to access the data selection, fusion tools and verification tools - In the header, the user is able to approve or reject the fusion result	GUI Technology: HTML Single view layout: - Axial view is displayed by default - The user is able to switch to coronal and sagittal views - A toolbar is available on the right side to access the data selection, fusion tools and verification tools - In the header, the user is able to approve or reject the fusion result
Region of Interest	The co-registration can be restricted to a specific region of interest of the datasets by means of a box.	The co-registration can be restricted to a specific region of interest of the datasets by means of a box and an ellipsoid.
Rigid co- registration, rigid image fusion	Automatic rigid co-registration (rigid fusion) of 3D DICOM image data: CT (incl. low-dose CT), MR, OT (e.g. Perfusion), MR-DTI, MR-BOLD, NM/PET, US (ultrasound). Manual rigid co-registration (rigid fusion) of other (unknown) 3D DICOM image data.	Automatic rigid co-registration (rigid fusion) of 3D DICOM image data: CT (incl. low-dose CT), MR, OT (e.g. Perfusion), MR-DWI, MR-BOLD, NM/PET, US (ultrasound). Manual rigid co-registration (rigid fusion) of other (unknown) 3D DICOM image data.
Deformable Registration / Elastic Image Fusion: Distortion	Distortion Correction for cranial image data: CT-MR, MR-MR (incl. DTI)	Distortion Correction for cranial image data: CT- MR, MR-MR (incl. DTI)

Topic/ Feature	Primary Predicate Device (Brainlab Elements Image Fusion 4.5 (K223106))	Subject Device (Brainlab Elements Image Fusion 5.0)
Correction Cranial		
Deformable Registration / Elastic Image Fusion: Curvature Correction Spine	Curvature Correction for spinal image data: CT-CT, CT-MR (incl. low-dose CT scanner)	Curvature Correction for spinal image data: CT- CT, CT-MR (incl. low-dose CT scanner)
Deformable Registration / Elastic Image Fusion: Virtual iMRI Cranial	Calculation of a brain shift (“virtual iMRI”) simulation, based on intra-operative CT or MR image data, offering a simulation of pre-operative planning data within an intra-operative situation. The intra-operative resection cavity is grayed out in the simulation result. With the acceptance of a Virtual iMRI simulation, the original variant of an image set and existing co-registrations are removed from the current data selection	Calculation of a brain shift (“virtual iMRI”) simulation, based on intra-operative CT, MR, or US image data, offering a simulation of pre-operative planning data within an intra-operative situation. The intra-operative resection cavity is blacked out in the simulation result. Automatic scrolling is activated as soon as a simulation result is available. With the acceptance of a Virtual iMRI simulation, the original variant of an image set and existing co-registrations are kept in the current data selection
Verification Tools: Spy Glass	A spy glass superimposes the first image set in a frame over the second image set.	A spy glass superimposes the first image set in a frame over the second image set.
Verification Tools: Blending	Blending allows to view structures in two image sets at the same time by using complimentary colors to distinguish both image sets.	Blending allows to view structures in two image sets at the same time by using complimentary colors to distinguish both image sets.
Verification Tools: Spy Glass for Deformable Co- Registrations	For deformable registrations the spy glass feature is combined with a heatmap and a deformation grid, highlighting areas where the deformed image set has been (locally) modified and giving a global impression about the deformation	For deformable registrations the spy glass feature is combined with a heatmap and a deformation grid, highlighting areas where the deformed image set has been (locally) modified and giving a global impression about the deformation
Verification Tools: Blending for deformable co-registrations	For deformable registrations the blending feature is combined with a deformation grid, which gives a global impression about the deformation while its colored grid lines highlight areas where the deformed image set has been (locally) modified.	For deformable registrations the blending feature is combined with a deformation grid, which gives a global impression about the deformation while its colored grid lines highlight areas where the deformed image set has been (locally) modified. For Virtual iMRI with Ultrasound, the weather radar, a specific type of blending with distinct colors, has been implemented.

Topic/ Feature	Primary Predicate Device (Brainlab Elements Image Fusion 4.5 (K223106))	Subject Device (Brainlab Elements Image Fusion 5.0)
Contrast Clearance Analysis	Calculation of treatment response assessment maps for the analysis of contrast agent clearance.	Calculation of treatment response assessment maps for the analysis of contrast agent clearance

Image Fusion Angio 1.0.2

Topic/ Feature	Primary Predicate Device (Brainlab Elements Image Angio 1.0.2 (K223106))	Subject Device (Brainlab Elements Image Fusion Angio 1.0.2)
Indications for use	Brainlab Elements Image Fusion Angio is a software application that is intended to be used for the co-registration of cerebrovascular image data. The device itself does not have clinical indications.	Brainlab Elements Image Fusion Angio is a software application that is intended to be used for the co-registration of cerebrovascular image data. It is not intended for diagnostic purposes. Brainlab Elements Image Fusion Angio is indicated for planning of cranial surgical treatments and preplanning of cranial radiotherapy treatments.
Operator profile	The intended users are medical professionals. Typical users are: - Image Guided Surgery (IGS): Neurosurgeons, Ear-Nose-Throat (ENT) surgeons and Cranio-Maxillofacial (CMF) surgeons including their assistants Radiotherapy (RT): medical professionals who perform radiation treatment planning (medical physicists, radiation oncologists, dosimetrists, physicians, etc.)	The intended users are medical professionals. Typical users are: - Image Guided Surgery (IGS): Surgeons trained in the areas of neurosurgery including their assistants. Radiotherapy (RT): medical professionals who perform radiation treatment planning (medical physicists, radiation oncologists, dosimetrists, physicians, etc.).
Computer hardware requirements	Brainlab Elements can be used on hardware that fulfills the defined minimum requirements: - Operating System: Windows 8.1 64bit - Minimum 4 logical cores - Minimum RAM: 6 GB - Graphics: Direct X compatible Display Resolution: 1920 x 1080 (Full HD)	All platforms which fulfill the minimum requirements: IGS Workstation: - Operating system: Windows 10 - minimum 4 physical cores - RAM: 8 GB - Graphics: DirectX 12 compatible - Display resolution: 1920 x 1080 (Full HD) IGS Server / Virtual Machine - Operating System: Windows Server 2016 - RAM 16GB RT Workstation: - Operating System: Windows 10 - minimum 6 physical cores

Topic/ Feature	Primary Predicate Device (Brainlab Elements Image Angio 1.0.2 (K223106))	Subject Device (Brainlab Elements Image Fusion Angio 1.0.2)
		- RAM: 48GB - Graphics card: DirectX 12 - Display resolution: 1920 x 1080 (Full HD) RT Server / Virtual Machine - Operating System: Windows Server 2016 - minimum 12 physical cores - RAM: 64GB (+16GB per additional user)

Contouring 5.0

Topic/ Feature	Primary Predicate Device (Brainlab Elements Contouring 4.5 (K223106))	Subject Device (Brainlab Elements Contouring 5.0)
Indications for use	Brainlab Elements Contouring provides an interface with tools and views to outline, refine, combine and manipulate structures in patient image data. The generated 3D structures are not intended to create physical replicas used for diagnostic purposes. The device itself does not have clinical indications	Brainlab Elements Contouring provides an interface with tools and views to outline, refine, combine and manipulate structures in patient image data. It is not intended for diagnostic purposes. Brainlab Elements Contouring is indicated for planning of cranial and extracranial surgical treatments and preplanning of cranial and extracranial radiotherapy treatments.
Patient population	In general, there are no demographic, regional or cultural limitations for patients. It is up to the user to decide if the system shall be used to assist a certain procedure.	There are no demographic, regional or cultural limitations for patients.
Conditions of use	The system shall be used in a hospital office environment or rooms appropriate for surgical interventions or radiotherapy planning.	The system shall be used in a hospital office environment or rooms appropriate for surgical interventions or radiotherapy planning.
Operator profile	The intended users are medical professionals. Typical users are: - Image Guided Surgery (IGS): Neurosurgeons, Ear-Nose-Throat (ENT) surgeons and Cranio-Maxillofacial (CMF) surgeons including their assistants Radiotherapy (RT): medical professionals who perform radiation treatment planning	The intended users are medical professionals. Typical users are: - Image Guided Surgery (IGS): Surgeons trained in the areas of neurosurgery, spine and trauma surgery, ear-nose-throat (ENT) surgery and craniomaxillofacial (CMF) surgery including their assistants. Radiotherapy (RT): medical professionals who perform radiation treatment planning (medical

Topic/ Feature	Primary Predicate Device (Brainlab Elements Contouring 4.5 (K223106))	Subject Device (Brainlab Elements Contouring 5.0)
	(medical physicists, radiation oncologists, dosimetrists, physicians, etc.)	physicists, radiation oncologists, dosimetrists, physicians, etc.).
Computer hardware requirements	Brainlab Elements can be used on hardware that fulfills the defined minimum requirements: - Operating System: Windows 8.1 64bit - Minimum 4 logical cores - Minimum RAM: 6 GB - Graphics: Direct X compatible - Display Resolution: 1920 x 1080 (Full HD)	IGS Workstation: - Operating system: Windows 10 - minimum 4 physical cores - RAM: 8 GB - Graphics: DirectX 12 compatible - Display resolution: 1920 x 1080 (Full HD) IGS Server / Virtual Machine: - Operating System: Windows Server 2016 - RAM 16GB RT Workstation: - Operating System: Windows 10 - minimum 6 physical cores - RAM: 48GB - Graphics card: DirectX 12 - Display resolution: 1920 x 1080 (Full HD) RT Server / Virtual Machine: - Operating System: Windows Server 2016 - minimum 12 physical cores - RAM: 64GB (+16GB per additional user) If Anatomical Patient Model 1.1 (APM 1.1) shall be used, the following minimum requirements have to be fulfilled: - CPU: 12 (virtual) cores - RAM: 24 GB If the Cranial Tumor Segmentation feature of Contouring shall be supported, the following GPU requirements have to be fulfilled: - GPU vRAM: 8GB - GPU driver: $\geq 472.5^*$ (*cuda 11.8 tool kit compatibility) - CUDA compute capability: > 5.2
Operating system	- Windows Server 2012 R2 / 2016 / 2019 - Windows 8.1 - Windows 10	- Windows Server 2016 - Windows Server 2019 - Windows Server 2022 - Windows 10
Input Devices	Touch and mouse/keyboard control	Touch and mouse/keyboard control
Automatic Segmentation: Cranial Tumors	Not available	Cranial tumors are auto-segmented as 3D objects in image sets with supported modality

Topic/ Feature	Primary Predicate Device (Brainlab Elements Contouring 4.5 (K223106))	Subject Device (Brainlab Elements Contouring 5.0)
		(MR-t1 contrast enhanced) by means of a machine learning algorithm. Similar to other anatomical structures the user can review, adjust or reject the tumor object.
Automatic Segmentation: Anatomical objects	The application automatically creates segmentation objects by using an atlas-based automatic segmentation of the brain. Workflow-relevant segmentation objects are created automatically or, if required by a workflow, on a single button click.	The application automatically creates segmentation objects by using an atlas-based automatic segmentation of the brain. Workflow-relevant segmentation objects are created automatically or, if required by a workflow, on a single button click. In addition auto-segmented objects can optionally also be retrieved via the Anatomical Patient Model 1.1 by means of a setting.
Landmark Support	The application can display annotations. Annotations cannot be changed or removed.	The application can be used to display, define and edit annotations and registration markers.
Visualization: Blending Mode	Not available	The application can be used to display two fused images within one view. A slider is available to blend from one image to another.
Abnormity Detection	If an anomaly is detected, the application automatically focuses the views on this anomaly. Anomaly detection is either the result of the atlas-based automatic segmentation of the brain or, on platforms with appropriate GPUs, the result of a machine learning algorithm. The GPU requirements are listed in the UniversalAtlasPerformer Test Plan ML Abnormity Detection.	If an anomaly is detected, the application automatically focuses the views on this anomaly. Anomaly detection is either the result of the atlas-based automatic segmentation of the brain or, on platforms with appropriate GPUs, the result of a machine learning algorithm. The GPU requirements are listed in the UniversalAtlasPerformer Test Plan ML Abnormity Detection.
Automatic Segmentation: Anatomical objects – Segmentation Templates	The application also offers the possibility to define customized segmentation templates.	The application also offers the possibility to define customized segmentation templates.
Automatic Segmentation – Resection Cavity	Not available	A resection cavity object can be segmented automatically based on ultrasound image data. Similar to other anatomical or tumor objects, the user can review, adjust or reject the resection cavity object.

Topic/ Feature	Primary Predicate Device (Brainlab Elements Contouring 4.5 (K223106))	Subject Device (Brainlab Elements Contouring 5.0)
Manual Creation and Refinement of Objects	The application provides tools for manual creation and refinement of segmentation objects: - SmartShaper - Brush 3D / Erase 3D - Brush 2D / Erase 2D - Smart Brush	The application provides tools for manual creation and refinement of segmentation objects: - SmartShaper - Brush 3D / Erase 3D - Brush 2D / Erase 2D - Smart Brush
Multi-Modal Drawing	SmartBrush can be used for multi-modal drawing, i.e. two images are used for drawing in a multi-modal view layout.	SmartBrush can be used for multi-modal drawing, i.e. two images are used for drawing in the newly introduced blending view when Blending Mode is active.
Manual Creation of Objects: Threshold Segmentation	The application provides a Threshold Segmentation tool to create objects. Threshold Segmentation supports MR, CT, PET, US and Contrast Clearance Analysis (CCA) data. The size of a geometric ROI during Threshold Segmentation can be adjusted.	The application provides a Threshold Segmentation tool to create objects. Threshold Segmentation supports MR, CT, PET, US, Contrast Clearance Analysis (CCA) and RGB data. The size of a geometric ROI during Threshold Segmentation can be adjusted and it can be rotated.
Object Manipulation	The application provides tools for the manipulation of objects: - Copy - Margins - Smoothing - Mirroring - Shifting and Rotation - Splitting - Automatic Object Fitting	The application provides tools for the manipulation of objects: - Copy - Margins - Smoothing - Mirroring - Shifting and Rotation - Splitting - Automatic Object Fitting
Logical Operations	The application provides tools for logical operations on objects: - Union - Subtract - Intersect	The application provides tools for logical operations on objects: - Union - Subtract - Intersect
Object Review	The application provides functionality to control the review state of objects. In case of leaving the application with unreviewed semi-automatically or automatically created objects, a message is displayed.	The application provides functionality to control the review state of objects. In case of leaving the application with unreviewed semi-automatically or automatically created objects, a message is displayed.
Volumetric Report	It is possible to create a printable report for an object in a platform independent file	It is possible to create a printable report for an object in a platform independent file format, giving information about the object's volume, its diameter according to the

Topic/ Feature	Primary Predicate Device (Brainlab Elements Contouring 4.5 (K223106))	Subject Device (Brainlab Elements Contouring 5.0)
	format, giving information about the object's volume, its diameter according to the Macdonald criteria and its response evaluation criteria (RECIST). If multiple objects are selected and a volumetric report is created, it contains information about all objects.	Macdonald criteria and its response evaluation criteria (RECIST). If multiple objects are selected and a volumetric report is created, it contains information about all objects.
Drawing Tools for the Spine SRS Use Case	The application calculates a clinical target volume (CTV) and a cropped organ at risk (OAR) object based on a manually outlined Gross Tumor (GTV). CTV and OAR objects can be created upon button click, based on a guideline published by Cox et al. (2012) when adding a GTV outline. In a user session, multiple combinations of GTV, CTV and OAR can be created.	The application calculates a clinical target volume (CTV) and a cropped organ at risk (OAR) object based on a manually outlined Gross Tumor (GTV). CTV and OAR objects can be created upon button click, based on a guideline published by Cox et al. (2012) when adding a GTV outline. In a user session, multiple combinations of GTV, CTV and OAR can be created.
Drawing Tools for the Angio Use Case	The application offers outlining features for cerebrovascular diseases. These features shall be supported by utilization of angiographic image data (i. e. 2D DSA) and projective fusions written by Image Fusion Angio.	The application offers outlining features for cerebrovascular diseases. These features shall be supported by utilization of angiographic image data (i. e. 2D DSA) and projective fusions written by Image Fusion Angio.

Fibertracking 3.0

Topic/ Feature	Primary Predicate Device (Brainlab Elements Fibertracking 2.0 (K223106))	Subject Device (Brainlab Elements Fibertracking 3.0)
Indications for use	Brainlab Elements Fibertracking is an application for the processing and visualization of cranial white matter tracts based on Diffusion Tensor Imaging (DTI) data for use in treatment planning procedures.	Brainlab Elements Fibertracking is an application for the processing and visualization of cranial white matter tracts based on diffusion weighted imaging (DWI) data for use in treatment planning procedures. It is not intended for diagnostic purposes. Brainlab Elements Fibertracking

Topic/ Feature	Primary Predicate Device (Brainlab Elements Fibertracking 2.0 (K223106))	Subject Device (Brainlab Elements Fibertracking 3.0)
	The device itself does not have clinical indications.	is indicated for planning of cranial surgical treatments and preplanning of cranial radiotherapy treatments.
Operator profile	The intended users are medical professionals. Typical users are: - Image Guided Surgery (IGS): Neurosurgeons, Ear-Nose-Throat (ENT) surgeons and Cranio-Maxillofacial (CMF) surgeons including their assistants Radiotherapy (RT): medical professionals who perform radiation treatment planning (medical physicists, radiation oncologists, dosimetrists, physicians, etc.)	The intended users are medical professionals. Typical users are: - Image Guided Surgery (IGS): Surgeons trained in the areas of neurosurgery including their assistants. Radiotherapy (RT): medical professionals who perform radiation treatment planning (medical physicists, radiation oncologists, dosimetrists, physicians, etc.).
Patient population	In general, there are no demographic, regional or cultural limitations for patients. It is up to the user to decide if the system shall be used to assist a certain procedure.	There are no demographic, regional or cultural limitations for patients.
Conditions of use	The system shall be used in a hospital office environment or rooms appropriate for surgical interventions or radiotherapy planning.	The system shall be used in a hospital office environment or rooms appropriate for surgical interventions or radiotherapy planning.
Computer hardware requirements	Brainlab Elements can be used on hardware that fulfills the defined minimum requirements: - Operating System: Windows 8.1 64bit - Minimum 4 logical cores - Minimum RAM: 6 GB - Graphics: Direct X compatible - Display Resolution: 1920 x 1080 (Full HD)	IGS Workstation: - Operating system: Windows 10 - minimum 4 physical cores - RAM: 8 GB - Graphics: DirectX 12 compatible - Display resolution: 1920 x 1080 (Full HD) IGS Server / Virtual Machine: - Operating System: Windows Server 2016 - RAM 16GB RT Workstation: - Operating System: Windows 10 - minimum 6 physical cores - RAM: 48GB - Graphics card: DirectX 12 - Display resolution: 1920 x 1080 (Full HD) RT Server / Virtual Machine: - Operating System: Windows Server 2016 - minimum 12 physical cores - RAM: 64GB (+16GB per additional user)
Operating system	- Windows Server 2012 R2 / 2016 / 2019 - Windows 8.1 - Windows 10	- Windows Server 2016 - Windows Server 2019 - Windows Server 2022 - Windows 10

Topic/ Feature	Primary Predicate Device (Brainlab Elements Fibertracking 2.0 (K223106))	Subject Device (Brainlab Elements Fibertracking 3.0)
Preprocessing	Automatic DTI Preprocessing (by using component DTI Preprocessing Performer 2.0): ☐ Eddy Current Correction, ☐ Motion Correction, ☐ B-Vector Realignment, ☐ Denoising, ☐ Regularization	Automatic DWI Preprocessing (by using component DWI Performer 3.0): ☐ Eddy Current Correction, ☐ Motion Correction, ☐ B-Vector Realignment, ☐ Denoising, ☐ Regularization ☐ Creation of ODF field for HARDI data
Define seed regions of interest	Application offers the following possibilities to define seed ROIs for Fibertracking (ROI based Fibertracking): ☐ Manual 3D Object (as include or exclude ROI) ☐ Existing 3D Object(s), e.g. by using ROI templates ☐ "Interactive Tracking" to use a dynamic ROI to generate fiber bundles on the fly	Application offers the following possibilities to define seed ROIs for Fibertracking (ROI based Fibertracking): ☐ Manual 3D Object (as include or exclude ROI) ☐ Existing 3D Object(s), e.g. by using ROI templates ☐ "Interactive Tracking" to use a dynamic ROI to generate fiber bundles on the fly
Algorithm configuration	Application offers three sliders to influence the algorithm's stopping criteria: ☐ Minimum FA Threshold (default: 0.2) ☐ Minimum Length (default: 80 mm) ☐ Maximum Angulation (default: 20°) See also topic "Toolarea".	Application offers sliders to influence the algorithm's stopping criteria: Deterministic tracking using DTI: ☐ Refinement (remove outliers, default: 10%) ☐ Minimum FA Threshold (default: 0.2) ☐ Maximum Angulation (default: 20°) Probabilistic tracking using CSD: ☐ Refinement (remove outliers, default: 10%) ☐ Amplitude (default: 0.1) ☐ Maximum Angulation (default: 20°)
Fibertracking algorithm	Deterministic Tracking Algorithm based on Combined Single/Dual Tensor Tracking Create 3D Objects from Fiber bundles	- Deterministic Tracking Algorithm based on Combined Single/Dual Tensor Tracking - Probabilistic tracking algorithm - Constrained Spherical Deconvolution tracking Create 3D Objects from Fiber bundles Reference Device K212397 StealthStation S8 Cranial v2.0: StealthStation S8 Cranial v2.0 offers a

Topic/ Feature	Primary Predicate Device (Brainlab Elements Fibertracking 2.0 (K223106))	Subject Device (Brainlab Elements Fibertracking 3.0)
		similar algorithm like the new probabilistic CSD tracking algorithm introduced with Fibertracking 3.0.
User Interface: Tool Area, description	□ Fiber bundles are accessible via toggle control and manageable via In-App DataSelection. □ ROIs are accessible in sidebar for drag&drop to fiber bundle. □ ROI-Brush accessible via toolarea. □ Other ROIs directly accessible via Sidebar. □ Parameter sliders area accessible in toolarea. □ “Refresh” accessible via toolarea. □ “Erase” accessible via toolarea □ “Objects Exclude” via Sidebar	□ Fiber bundles are accessible and manageable in sidebar control □ ROIs are accessible in sidebar for drag&drop to fiber bundle and selectable for editing. □ ROI-Brush accessible via toolarea. □ Other ROIs directly accessible via Sidebar. □ Parameter sliders area accessible in toolarea. □ “Refresh” accessible via sidebar. □ “Erase” accessible via toolarea. □ “Objects Exclude” via Sidebar □ Algorithm (DTI or CSD) can be selected in toolarea □ Measurement of coordinates and FA values available via toolarea

BOLD MRI Mapping 1.0

Topic/ Feature	Primary Predicate Device (Brainlab Elements BOLD MRI Mapping (K223106))	Subject Device (Brainlab Elements BOLD MRI Mapping 1.0)
Indications for use	Brainlab Elements BOLD MRI Mapping provides tools to analyze blood oxygen level dependent data (BOLD MRI Data) to visualize the activation signal and correlation between brain areas. The device itself does not have clinical indications	Brainlab Elements BOLD MRI Mapping provides tools to analyze blood oxygen level dependent data (BOLD MRI Data) to visualize the activation signal. It is not intended for diagnostic purposes. Brainlab Elements BOLD MRI Mapping is indicated for planning of cranial surgical treatments.
Operator profile	The intended users are medical professionals. Typical users are:	The intended users are medical professionals. Typical users are:

Topic/ Feature	Primary Predicate Device (Brainlab Elements BOLD MRI Mapping (K223106))	Subject Device (Brainlab Elements BOLD MRI Mapping 1.0)
	- Image Guided Surgery (IGS): Neurosurgeons, Ear-Nose-Throat (ENT) surgeons and CranioMaxillofacial (CMF) surgeons including their assistants - Radiotherapy (RT): medical professionals who perform radiation treatment planning (medical physicists, radiation oncologists, dosimetrists, physicians, etc.)	Image Guided Surgery (IGS): Surgeons trained in the area of neurosurgery including their assistants
Computer hardware requirements	Brainlab Elements can be used on hardware that fulfills the defined minimum requirements: - Operating System: Windows 8.1 64bit - Minimum 4 logical cores - Minimum RAM: 6 GB - Graphics: Direct X compatible Display Resolution: 1920 x 1080 (Full HD)	IGS Workstation: - Operating system: Windows 10 - minimum 4 physical cores - RAM: 8 GB - Graphics: DirectX 12 compatible - Display resolution: 1920 x 1080 (Full HD) IGS Server / Virtual Machine: - Operating System: Windows Server 2016 - RAM 16GB
Operating system	- Windows Server 2012 R2 / 2016 / 2019 - Windows 8.1 Windows 10	- Windows Server 2016 - Windows Server 2019 - Windows Server 2022 Windows 10

4. Performance Data

The Subject Device has been verified and validated according to Brainlab processes for product design and development. A high level explanation of the testing provided in this submission for the Subject Device is provided below.

Software verification:

Software verification has been performed on software level, verifying the software requirements through integration tests as well as GUI tests and Unit tests. An incremental test strategy has been set up for the release candidates for changes with limited scope. In this case, an impact analysis of the modifications was performed and tests to be performed are identified and planned correspondingly.

Usability Evaluation:

Image Fusion 5.0:

Formative and Summative Usability was carried out for Image Fusion 5.0. The main goal was the calculation of vMRI images that can support further neurosurgical procedures. Compared to Image Fusion 4.5, users can now additionally calculate a vMRI based on an intraoperative Ultrasound. The page layout and tools for verification of a fusion remain similar to predecessor versions.

Fibertracking 3.0:

The changes did not affect primary operating functions or any use related risks. No Summative Usability Evaluation was performed. Four rounds of formative evaluations of the changes were performed with a total of 23 users. Different rounds focused on various aspects such as general use of Fibertracking in a clinical workflow and experiences with the predecessor devices, new GUI components, review of the final concepts for the user interface.

Accuracy Tests:

Image Fusion 5.0:

This test was to validate that Elements Virtual iMRI Cranial can be applied to cranial MR and intraoperative MR/CT/US image data related to image guided surgery approaches to compensate for surgery-related brain shift during resection. Tests were carried out using retrospective image data sets.

Fibertracking 3.0:

To validate if Fibertracking allows to visualize cranial white matter structures such as motoric, language and visual tracts based on state of the art approaches for Fibertracking as Constrained Spherical Deconvolution (CSD) and probabilistic tracking.

AI/ML Performance Tests for Contouring 5.0:

To validate Elements AI Tumor Segmentation can be used to semi-automatically segment supported cranial tumors (metastases, meningiomas, cranial and paraspinal nerve tumors, gliomas, glioneuronal and neuronal tumors) on 3D medical Contrast Enhanced T1-weighted MR images powered by the Anatomical Patient Model. The approach used was quantitative validation – comparison to ground-truth annotations.

Validation Summary:

Elements AI Tumor Segmentation algorithm was tested on de-identified 3D CE-T1 MR images from 412 patients (595 scans, 1878 annotations) from multiple clinical sites in the US and Europe. All data was acquired from adult patients with one or multiple contrast-enhancing tumors. The pool contained homogenous distribution by gender (female/male) and a diversity of ethnicity groups being White/Black/Latino/Asian with the highest representation. Most of data were obtained in patients who underwent stereotactic radiosurgery using a diverse acquisition of MR protocols (mainly containing 1.5T/3T MRI scans acquired in axial scan orientation). The test data pools also included control patients without cranial tumors. ¼ of the test pool corresponded to data from three independent sites in USA. The validation was conducted quantitatively by comparing the (manual) ground-truth segmentations, the so-called annotations with the respective automatically-created segmentations. The annotations involved external/independent annotator team including US radiologists and non US radiologists which followed a well-defined data curation process.

In accordance with best practices, measurement of recall and precision and object-specific DICE coefficient were performed to assess the similarity between both segmentations. Acceptance criteria were Dice ≥ 0.7 , Recall ≥ 0.8 and Precision ≥ 0.8 for the lower bound of the respective 95 % confidence intervals. Successful validation has been completed based on images containing up to 30 cranial metastases, each showing a diameter of at least 3 mm, and images with primary cranial tumors that are at least 10 mm in diameter (for meningioma, cranial/paraspinal nerve tumors, gliomas, glioneuronal and neuronal tumors). See performance metrics in Table 1.

Table 1 Summary of test statistics

Diagnostic Characteristics	Mean Dice	Mean Precision	Mean Recall
All	0.75	0.86	0.85
Metastases to the CNS	0.74	0.85	0.84

Meningiomas	0.76	0.89	0.90
Cranial and paraspinal nerve tumors	0.89	0.97	0.97
Gliomas and glio-/neuronal tumors	0.81	0.95	0.85

5. Conclusion

The comparison of the Subject Device with the predicate device shows that Brainlab Elements 7.0 has similar functionality, intended use and technological characteristics as the predicate device. Based on the comparison to the predicate and the performance testing conducted, the Subject Device is considered substantially equivalent to the predicate device.